

M08 MODULE TEST - STATISTICS TRANSFER CHECK

RC2 | Non-gating | Fresh problems | STAT1-STAT9 coverage

Status: non-gating transfer diagnostic. It does not add a prerequisite and does not change the canonical route STAT1-STAT9 -> M-STAT -> Gate A.

Attempt conditions

- Save a genuine first attempt before opening the protected review. Do not use the learner-book review pages, Gate reviews or AI-generated solutions during the first attempt.
- A calculator/spreadsheet may be used for arithmetic. The reasoning, units, assumptions and interpretation must still be written in your own words.
- Equivalent correct methods are accepted. A formula result without interpretation is incomplete.
- This diagnostic can identify repair areas but can never unlock Gate B or Gate C.

Part A - Description and distribution (4 points)

A1. Delivery minutes: 18, 20, 21, 22, 23, 24, 25, 26, 29, 82. Compute count, mean, median and range. Explain what the 82-minute observation does to mean vs median and whether it should be deleted automatically.

A2. A metric has mean 50 and SD 8. An observation is 66. Compute its z-score and state what that does - and does not - tell you.

Part B - Sampling, SE and confidence intervals (5 points)

B1. A product team wants to generalize to all monthly active users but surveys only users who opened the app during the last 7 days. It gets 5,000 responses. Identify the target population, the sampling frame and the main bias risk. Does large n repair that bias?

B2. A sample has mean 72, sample SD 18 and $n=144$. Compute SE and an approximate 95% CI using mean $\pm 1.96 \times SE$. Interpret the interval using the repeated-procedure idea and name one limitation.

B3. If n rises from 144 to 576 with the same SD 18, compute the new SE and explain the precision change.

Part C - Hypothesis test, effect size and business threshold (6 points)

Fresh experiment: Control 1,200 users / 120 conversions; Treatment 1,200 / 150 conversions. $\alpha=0.05$. Minimum worthwhile effect=2.0pp. Guardrail: support-contact increase must be $\leq 1.0pp$. Support contacts: Control 72; Treatment 78.

C1. Compute both conversion rates, absolute lift in percentage points and relative lift %. Then compute/verify the two-proportion z statistic, two-sided p-value and approximate 95% CI for the rate difference.

C2. Compute support-contact rates and the pp change. State whether the observed effect clears the 2.0pp point-estimate threshold, whether the 95% CI proves an effect of at least 2.0pp, whether the guardrail is breached, and the appropriate decision language.

C3. In one sentence, explain why $p=0.053$ is not a 5.3% probability that H_0 is true.

Part D - Correlation and causal limits (2 points)

An observational analysis reports $r=0.84$ between employee training hours and performance score. Write one defensible association statement, name two plausible alternative explanations/confounders, and state one stronger design for causal identification.

Part E - A/B design discipline (3 points)

Before seeing results, outline a signup-flow experiment: unit of randomization, eligibility, control/treatment, one primary metric, one guardrail, planned duration/stopping rule, minimum worthwhile effect, sample-ratio/data-quality check and final decision rule.

Score interpretation

16-20: strong transfer. **12-15:** targeted repair only. **0-11:** return to the named weak competencies, not the entire module. Regardless of score, the canonical Gate route remains unchanged.